UBISS 2026: Minimalism in Robotics

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Instructors



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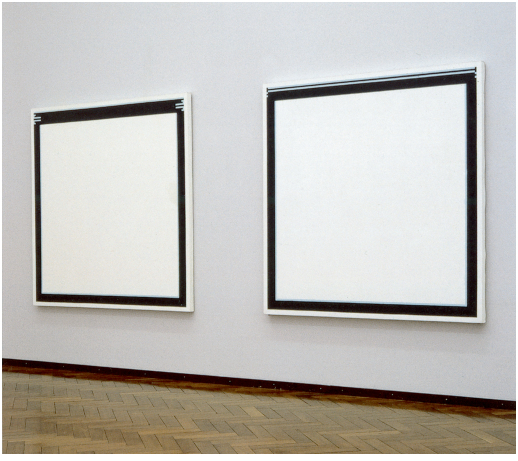
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Maastricht University



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What is Minimalism?

An Aesthetic



Paintings in the “Korean” series by Jo Baer

A Useful Constraint for Design

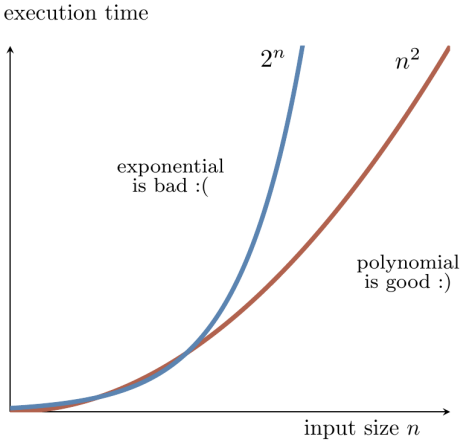


Source: Eva & Franco Mattes



Source: gatsby.bot

A Lens for Analysis



Source: qubit.guide

Why Minimalism in Robotics?

Organizing Robotics as a Synthetic Science

For robotics to develop as a science[1], we need theory that can:

- distinguish between and relate **tasks**,

[1] "What Is Robotics? Why Do We Need It and How Can We Get It?" Dan Koditschek 2021

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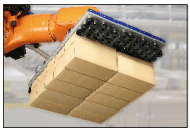
- distinguish between and relate **tasks**,
- **architectures** for accomplishing them, and
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We need to be able to compare concepts within and across these domains.

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Robot Design is not a Simple Optimization Problem

- Many feasible designs per task
- Resource trade-offs
- Highly coupled complex systems
- Messy regions of infeasibility



Vacuum Gripper



Pneumatic Gripper



Hydraulic Gripper

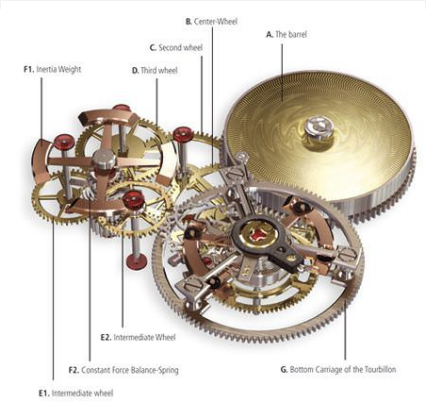


Servo-Electric Gripper

Source: Universal Robots

Benefits of Abstraction

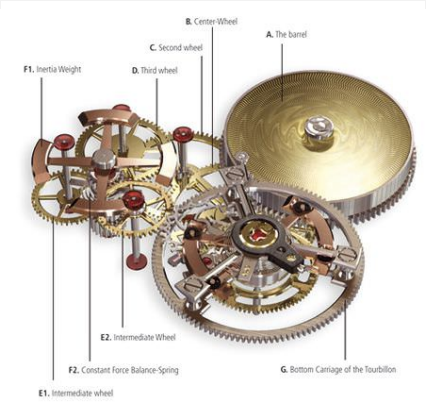
- Modularity and composition



Source: MasterHorologer

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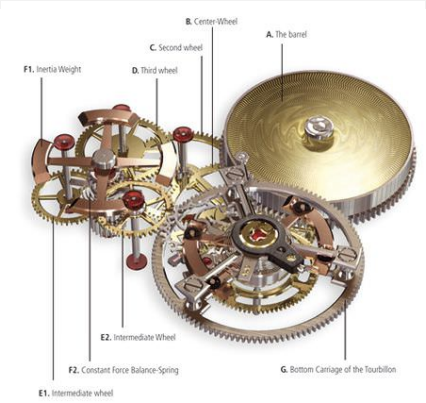
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- Comparing and classifying problems (reducing one problem to another)



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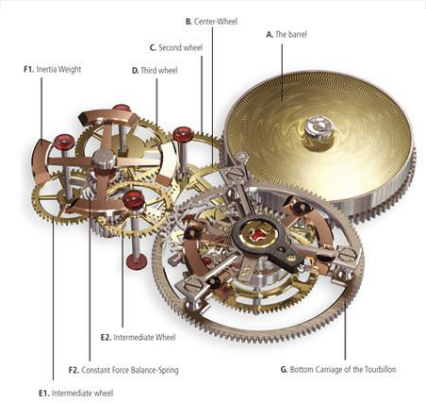
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- Comparing and classifying problems (reducing one problem to another)
- Computational / information complexity results



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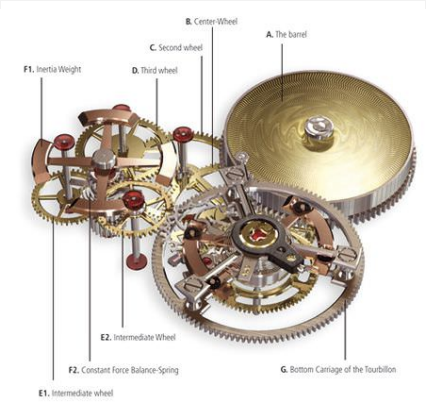
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- Impossibility results: boundary of feasibility



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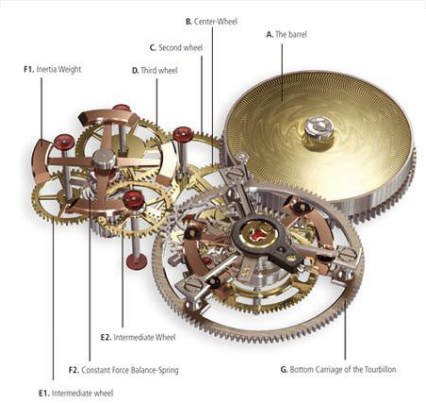
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Minimalism helps us make *better abstractions* by intentionally removing *unnecessary details* from robotics problems!

Other Benefits of Minimalism

- Resource efficiency

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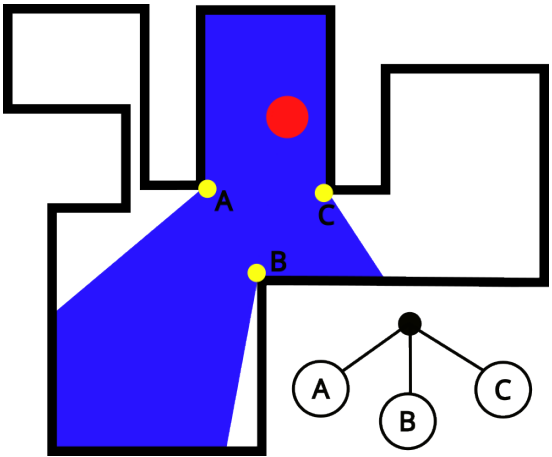
- Resource efficiency
- Ease of system debugging and verification

Other Benefits of Minimalism

- Resource efficiency
- Ease of system debugging and verification
- Supply chain robustness

How to Do Minimalism in Robotics?

Example: Gap Navigation Trees



See "Gap navigation trees: A minimal representation for visibility-based tasks" in WAFR 2005 by S.M. LaValle et. al.

Back to the Bigger Picture

Quest for Understanding?	Considerations of Use?	
	No	**Yes**
Yes	Basic Research	Use-Inspired Basic Research
No	(Quadrant of Confusion)	Applied Research

Pasteur's Quadrant classification of research, origin Donald E. Stokes

Back to the Bigger Picture

No matter what kind of roboticist you are, minimalism is a useful paradigm for thinking about robot design!

Thank you!

